

PROBLEMS CLASS 4: SOLUTIONS

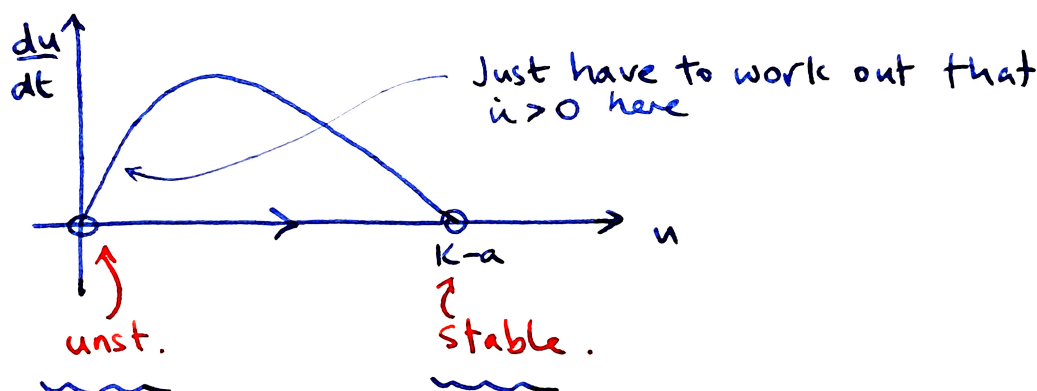
1. Travelling waves in skin repair (Exam 2022, section B)

(a) The homogeneous equation is

$$\frac{du}{dt} = -u \log \left(\frac{u+a}{K} \right).$$

The equilibria are at $u = 0$ and $u = K - a$.

Their stability can be determined graphically:



or spotting that for $0 < u \ll 1$, $du/dt > 0$ because $\log(a/K) < 0$ since $a < K$. So $u = 0$ is unstable and since the stability of 1D systems alternates, $u = K - a$ must be stable.

(b) Description of the model:

- diffusion with a growth term, which has a carrying capacity $u = K - a$,
- similar to FK (see graph above) in shape and having a carrying capacity, but FK is logistic, ' $+u(1-u)$ '.

(c) Argument of the form:

- Expect the stable equilibrium, $u = K - a$, to expand, so the wave will travel away from this.
- A wave of the form $z = x - ct$ moves to the *right*.
- Therefore pick

$$u(-\infty) = K - a, \quad u(\infty) = 0.$$

(d) Substituting in $z = x - ct$, we have

$$\frac{\partial u}{\partial t} = -c \frac{du}{dz} \quad \text{and} \quad \frac{\partial^2 u}{\partial x^2} = \frac{d^2 u}{dz^2},$$

so

$$-c \frac{du}{dz} = \frac{d^2 u}{dz^2} - u \log \left(\frac{u+a}{K} \right).$$

To perform the linear stability analysis, let $v = du/dz$ then

$$\begin{aligned}\frac{du}{dz} &= v, \\ \frac{dv}{dz} &= -cv + u \log \left(\frac{u+a}{K} \right).\end{aligned}$$

The Jacobian is

$$\mathbf{J} = \begin{pmatrix} 0 & 1 \\ \log \left(\frac{u+a}{K} \right) + \frac{u}{u+a} & -c \end{pmatrix},$$

where the bottom-left entry is worked out as

$$\begin{aligned}\frac{d}{du} \left[u \log \left(\frac{u+a}{K} \right) \right] &= \log \left(\frac{u+a}{K} \right) + u \frac{K}{u+a} \frac{1}{K} \\ &= \log \left(\frac{u+a}{K} \right) + \frac{u}{u+a}.\end{aligned}$$

Evaluated at $(0,0)$, the Jacobian is

$$\mathbf{J} = \begin{pmatrix} 0 & 1 \\ \log \frac{a}{K} & -c \end{pmatrix}.$$

The eigenvalue equation is therefore

$$\begin{aligned}-\lambda(-\lambda - c) - \log \frac{a}{K} &= 0 \\ \implies \lambda^2 + c\lambda - \log \frac{a}{K} &= 0,\end{aligned}$$

and hence

$$\implies \lambda = \frac{1}{2} \left(-c \pm \sqrt{c^2 + 4 \underbrace{\log \frac{a}{K}}_{<0}} \right).$$

This is therefore a stable node (two negative, real eigenvalues) if

$$c^2 > -4 \log \frac{a}{K}, \tag{1}$$

else it is a spiral (complex eigenvalues).

The Jacobian, evaluated at $(K-a, 0)$, is

$$\mathbf{J} = \begin{pmatrix} 0 & 1 \\ \frac{K-a}{K} & -c \end{pmatrix},$$

and hence

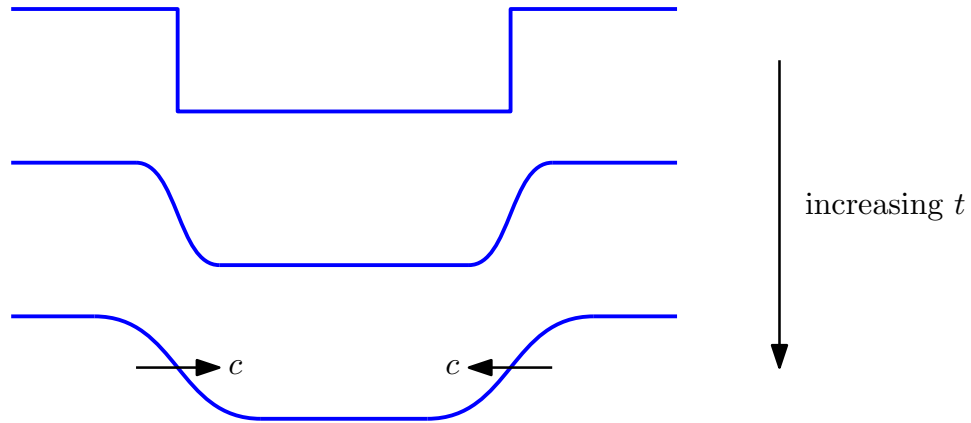
$$\implies \lambda = \frac{1}{2} \left(-c \pm \sqrt{c^2 + 4 \underbrace{\frac{K-a}{K}}_{>0}} \right).$$

The eigenvalues are real, with different sign. This is therefore a saddle point.

A spiral is not possible, as that would mean negative u , which is unphysical. We therefore conclude eq. (1) is true, and, further, since waves travel at their lowest possible speed,

$$c = 2\sqrt{\log \frac{K}{a}}.$$

(e) Expect $u(x, t)$ to develop like:



with a closure time of approximately

$$\frac{L/2}{c} = \frac{L}{4\sqrt{\log(K/a)}}.$$

2. Disease modelling with vaccination (Exam 2022, section B)

(a) We have that

$$\begin{aligned} \frac{dN}{dt} &= \frac{dS}{dt} + \frac{dI}{dt} + \frac{dR}{dt} \\ &= (\alpha R - \beta IS - \nu S) + (\beta IS - \gamma I) + (\gamma I - \alpha R + \nu S) \\ &= 0. \end{aligned}$$

Eliminating R , our system becomes

$$\begin{aligned} \frac{dS}{dt} &= \alpha(N - S - I) - \beta IS - \nu S, \\ \frac{dI}{dt} &= \beta IS - \gamma I. \end{aligned}$$

We can lose the equation for dR/dt since dS/dt and dI/dt are independent of R , and because

$$\frac{dR}{dt} = -\frac{dS}{dt} - \frac{dI}{dt}$$

anyway.

(b) Interpretation:

- α : rate at which immunity wears off; any unit that fits the dimension of time^{-1} ,
- β : infection rate; $\text{people}^{-1}\text{time}^{-1}$,
- γ : recovery rate (possibly death rate, but dubious since it is possible to leave the group R); time^{-1} .

(c) Example arguments [students should provide three]:

- no because there is no incubation period (as in an SEIR model),
- no because dead people cannot be reinfected

- no because vaccination doesn't necessarily stop infection, it may just reduce the rate of infection
- no because no spatial dependence
- yes because it captures the general shape of epidemic 'waves'

(d) (i) Equilibria are where

$$0 = \alpha(N - S - I) - \beta IS - \nu S, \quad (2)$$

$$0 = \beta IS - \gamma I. \quad (3)$$

Equation (3) gives us

$$I = 0 \quad \text{or} \quad S = \frac{\gamma}{\beta}.$$

If $I = 0$ then eq. (2) gives us

$$\begin{aligned} 0 &= \alpha(N - S) - \nu S \\ \implies \alpha N &= (\alpha + \nu)S \\ \implies S &= \frac{\alpha}{\alpha + \nu}N. \end{aligned}$$

And if $S = \gamma/\beta$ then

$$\begin{aligned} 0 &= \alpha \left(N - \frac{\gamma}{\beta} - I \right) - I\gamma - \frac{\nu\gamma}{\beta} \\ \implies \alpha N - \frac{\alpha\gamma}{\beta} - \frac{\gamma\nu}{\beta} &= I(\alpha + \gamma) \\ \implies \alpha N - (\alpha + \nu)\frac{\gamma}{\beta} &= I(\alpha + \gamma) \\ \implies I &= \frac{\alpha\beta N - (\alpha + \nu)\gamma}{\beta(\alpha + \gamma)}. \end{aligned}$$

The two equilibria are therefore

$$\left(\frac{\alpha N}{\alpha + \nu}, 0 \right) \quad \text{and} \quad \left(\frac{\gamma}{\beta}, \frac{\alpha\beta N - (\alpha + \nu)\gamma}{\beta(\alpha + \gamma)} \right) =: (S^*, I^*).$$

(ii) The Jacobian is

$$J = \begin{pmatrix} -\alpha - \beta I - \nu & -\alpha - \beta S \\ \beta I & \beta S - \gamma \end{pmatrix}.$$

Evaluated at $\left(\frac{\alpha N}{\alpha + \nu}, 0 \right)$, it is

$$\begin{pmatrix} -\alpha - \nu & -\alpha - \frac{\beta\alpha N}{\alpha + \nu} \\ 0 & \frac{\beta\alpha N}{\alpha + \nu} - \gamma \end{pmatrix}.$$

The eigenvalues are

$$-\alpha - \nu < 0 \quad \text{and} \quad \frac{\beta\alpha N}{\alpha + \nu} - \gamma.$$

So we can say the eigenvalue at $\left(\frac{\alpha N}{\alpha + \nu}, 0 \right)$ is unstable if $\frac{\beta\alpha N}{\alpha + \nu} > \gamma$.

For the endemic equilibrium, (S^*, I^*) , use the trace and determinant method.

$$\begin{aligned}\text{tr } \mathbf{J} &= \alpha - \beta I^* - \nu + \beta S^* - \gamma \\ &= -\alpha - \beta I^* - \nu < 0.\end{aligned}$$

And the determinant, noting that the bottom-right entry of the Jacobian is 0, is

$$\det \mathbf{J} = 0 - (-\alpha - \gamma)(I^* \beta) \quad (4)$$

$$= (\alpha + \gamma)I^*\beta > 0. \quad (5)$$

Since $\text{tr } \mathbf{J} < 0$ and $\det \mathbf{J} > 0$, we conclude that (S^*, I^*) is always stable if it exists.

Further, (S^*, I^*) exists if

$$\alpha\beta N - (\alpha + \nu)\gamma > 0$$

$$\implies \left(\frac{\alpha}{\alpha + \nu}\right)\beta N > \gamma,$$

which is the same condition.

The disease is therefore eradicated if

$$R^* := \frac{\alpha}{\alpha + \nu} \frac{\beta N}{\gamma} < 1. \quad (6)$$

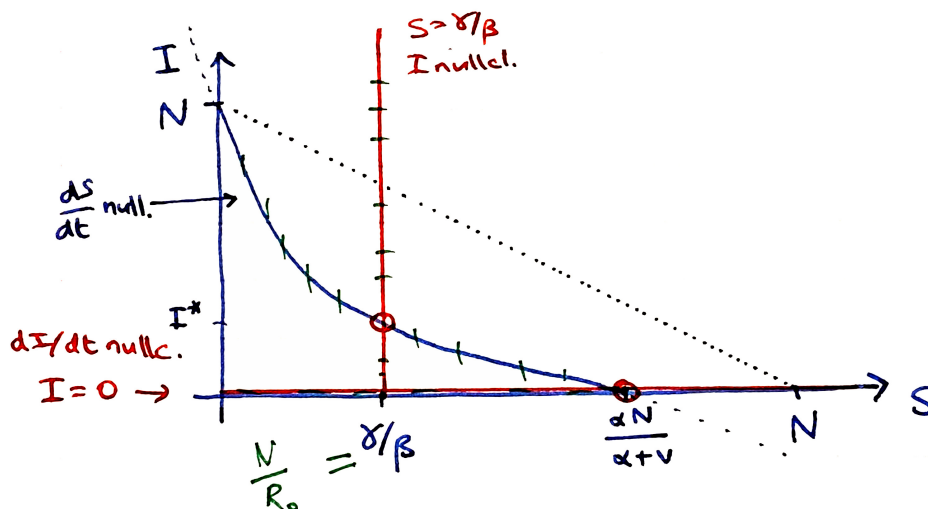
(iii) From eq. (6), we define

$$R_0 = \frac{\beta N}{\gamma},$$

and therefore have

$$\begin{aligned} \frac{\alpha}{\alpha + \nu} R_0 < 1 &\implies \alpha R_0 < \alpha + \nu \\ &\implies \alpha(R_0 - 1) < \nu. \end{aligned}$$

(e) Phase portrait:



The I nullclines are $I = 0$ and $S = \gamma/\beta$.

The S nullcline is where

$$\begin{aligned} 0 &= \alpha N - \alpha S - \alpha I - \beta IS - \nu S \\ \implies I(\alpha + \beta S) &= \alpha(N - S) - \nu S \\ \implies I &= \frac{\alpha N - (\alpha + \nu)S}{\alpha + \beta S}. \end{aligned}$$

This is awkward to plot but notice that

$$I(0) = N \quad \text{and} \quad I\left(\frac{\alpha N}{\alpha + \nu}\right) = 0.$$

You can also notice

- for small S , I looks linear,
- for large S , I looks constant,

which implies the shape in the picture.

As for how the existence of the endemic equilibrium depends on ν and R_0 :

- the equilibrium exists where the S and I nullclines cross,
- the I nullcline is at $\gamma/\beta = N/R_0$,
- changing R_0 shifts the I nullcline left and right,
- changing ν squeezes the S nullcline horizontally,
- it is possible to eradicate the disease by increasing ν or decreasing R_0 .